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day / date:

Design Analysis & Algorithm - Assignment 4

Q.1)

The Bellman-Ford algorithm normally relaxes all $|V|-1$ times, but we don't always need to do that. After each full pass over all edges, if none of the distances change, then all the shortest paths have already been found, because any improvement would have shown up during that pass. So we will keep a flag during every iteration & see it whenever a relaxation updates a vertex's distance. If after an entire pass the flag is still FALSE, that means no distance can be improved further, and then we can stop the algorithm early instead of completing all remaining passes. This works because once no edge can reduce it's endpoint's distance, every reachable vertex already has the correct shortest-path value. We will still do one final check to see if any edge can be relaxed again; if it can then that indicates a negative weight cycle. Otherwise, the algorithm finishes with the correct shortest paths.

```
BELLMAN FORD EARLY TERMINATION (G, w, s)
    //initialization
    for each vertex v in V[G]
        v.d = ∞
        v.π = nil
    s.d = 0
    // relax edges repeatedly until no update happens
    repeat
        changed = FALSE
        for each edge (u,v) in E[G]
            if v.d > u.d + w(u,v)
                v.d = u.d + w(u,v)
                v.π = u
                changed = TRUE
    until changed == FALSE
```

day / date:

// check for negative-weight cycles
for each edge (u,v) in E[G]
  if v.d > u.d + w(u,v)
    return "negative weight cycle detected"

return distances and predecessors

Q.2) $r \to s \to t \to x \to y \to z$

$d[r]=0$, $\pi[r]=nil$
$d[s]=\infty$, $\pi[s]=nil$
$d[t]=\infty$, $\pi[t]=nil$
$d[x]=\infty$, $\pi[x]=nil$
$d[y]=\infty$, $\pi[y]=nil$
$d[z]=\infty$, $\pi[z]=nil$

vertex	edge	updated d[] values	updated π[] values
r	r→s(5)	s=5 , d[s]=5	π[s]=r
	r→t(3)	t=3, d[t]=3	π[t]=r
		d[r]=0, d[x]=d[y]	
		d[z]=∞	
s	s→t(2)	3<5+2 , d[t]=3	
	s→x(6)	x=5+6=11 , d[x]=11	π[x]=s
t	t→x(7)	x=3+7=10 , d[x]=10	π[x]=t
	t→y(4)	y=3+4=7 , d[y]=7	π[y]=t
	t→z(2)	z=3+2=5 , d[z]=5	π[z]=t
x	x→y(-1)	7<10+(-1), d[y]=7	
	x→z(+1)	5<10+1 , d[z]=5	
y	y→z(-2)	5=7+(-2), d[z]=5	

↳ time complexity : O(V+E)

↳ final path:
$r \to t \to z$

↳ cost = 5

day / date:

Q.3)

↳ 's' as source vertex

vertex	D[s]	D[t]	D[x]	D[y]	D[z]	π[t]	π[x]	π[y]	π[z]	S
s	0	10,s	∞	5,s	∞	s	–	s	–	{s}
y	0	8,y	14,y		7,y	y	y	s	y	{s,y}
z	0	8,y	13,z			y	z	s	y	{s,y,z}
t	0		9,t			y	t	s	y	{s,y,z,t}
x	0	8	9	5	z	y	t	s	y	{s,y,z,t,x}

↳ 'z' as source vertex

vertex	D[z]	D[y]	D[x]	D[t]	D[s]	π[y]	π[x]	π[t]	π[s]	S
z	0	∞	6,z	∞	7,z	–	z	–	z	{z}
x	0	∞		∞	7,z	–	z	–	z	{z,x}
s	0	12,s		17,s		s	z	s	z	{z,x,s}
y	0			15,y		s	z	y	z	{z,x,s,y}
t	0	12	6	15	7	s	z	y	z	{z,x,s,y,t}

Q.4) $A_k[i,j] = \min\left[A^{k-1}[i,j],\ A^{k-1}[i,k] + A^{k-1}[k,j]\right]$

a) $A^0 = \begin{bmatrix} 0 & \infty & \infty & \infty & -1 & \infty \\ 1 & 0 & \infty & 2 & \infty & \infty \\ \infty & 2 & 0 & \infty & \infty & -8 \\ -4 & \infty & \infty & 0 & 3 & \infty \\ \infty & 7 & \infty & \infty & 0 & \infty \\ \infty & 5 & 10 & \infty & \infty & 0 \end{bmatrix}$ $A^1 = \begin{bmatrix} 0 & \infty & \infty & \infty & -1 & \infty \\ 1 & 0 & \infty & 2 & 0 & \infty \\ \infty & 2 & 0 & \infty & \infty & -8 \\ -4 & \infty & \infty & 0 & -5 & \infty \\ \infty & 7 & \infty & \infty & 0 & \infty \\ \infty & 5 & 10 & \infty & \infty & 0 \end{bmatrix}$

$A^2 = \begin{bmatrix} 0 & \infty & \infty & \infty & -1 & \infty \\ 1 & 0 & \infty & 2 & 0 & \infty \\ 3 & 2 & 0 & 4 & 2 & -8 \\ -4 & \infty & \infty & 0 & -5 & \infty \\ 8 & 7 & \infty & 9 & 0 & \infty \\ 6 & 5 & 10 & 7 & 5 & 0 \end{bmatrix}$ $A^3 = \begin{bmatrix} 0 & \infty & \infty & \infty & -1 & \infty \\ 1 & 0 & \infty & 2 & 0 & \infty \\ 3 & 2 & 0 & 4 & 2 & -8 \\ -4 & \infty & \infty & 0 & -5 & \infty \\ 8 & 7 & \infty & 9 & 0 & \infty \\ 6 & 5 & 10 & 7 & 5 & 0 \end{bmatrix}$

day / date:

$A^4 = \begin{bmatrix} 0 & \infty & \infty & \infty & -1 & \infty \\ -2 & 0 & \infty & 2 & -3 & \infty \\ 0 & 2 & 0 & 4 & -1 & -8 \\ -4 & \infty & \infty & 0 & -5 & \infty \\ 5 & 7 & \infty & 9 & 0 & \infty \\ 3 & 5 & 10 & 7 & 2 & 0 \end{bmatrix}$

$A^5 = \begin{bmatrix} 0 & 6 & \infty & 8 & -1 & \infty \\ -2 & 0 & \infty & 2 & -3 & \infty \\ 0 & 2 & 0 & 4 & -1 & -8 \\ -4 & 2 & \infty & 0 & -5 & \infty \\ 5 & 7 & \infty & 9 & 0 & \infty \\ 3 & 5 & 10 & 7 & 2 & 0 \end{bmatrix}$

$A^6 =$ final $\begin{bmatrix} 0 & 6 & \infty & 8 & -1 & \infty \\ -2 & 0 & \infty & 2 & -3 & \infty \\ -5 & -3 & 0 & -1 & -6 & -8 \\ -4 & 2 & \infty & 0 & -5 & \infty \\ 5 & 7 & \infty & 9 & 0 & \infty \\ 3 & 5 & 10 & 7 & 2 & 0 \end{bmatrix}$

b)

$A^0 = \begin{bmatrix} 0 & 3 & 8 & \infty & -4 \\ \infty & 0 & \infty & 1 & 7 \\ \infty & 4 & 0 & \infty & \infty \\ 2 & \infty & -5 & 0 & \infty \\ \infty & \infty & \infty & 6 & 0 \end{bmatrix}$

$A^1 = \begin{bmatrix} 0 & 3 & 8 & \infty & -4 \\ \infty & 0 & \infty & 1 & 7 \\ \infty & 4 & 0 & \infty & \infty \\ 2 & 5 & -5 & 0 & -2 \\ \infty & \infty & \infty & 6 & 0 \end{bmatrix}$

$A^2 = \begin{bmatrix} 0 & 3 & 8 & 4 & -4 \\ \infty & 0 & \infty & 1 & 7 \\ \infty & 4 & 0 & 5 & 11 \\ 2 & 5 & -5 & 0 & -2 \\ \infty & \infty & \infty & 6 & 0 \end{bmatrix}$

$A^3 = \begin{bmatrix} 0 & 3 & 8 & 4 & -4 \\ \infty & 0 & \infty & 1 & 7 \\ \infty & 4 & 0 & 5 & 11 \\ 2 & -1 & -5 & 0 & -2 \\ \infty & \infty & \infty & 6 & 0 \end{bmatrix}$

$A^4 = \begin{bmatrix} 0 & 3 & -1 & 4 & -4 \\ 3 & 0 & -4 & 1 & -1 \\ 7 & 4 & 0 & 5 & 3 \\ 2 & -1 & -5 & 0 & -2 \\ 8 & 5 & 1 & 6 & 0 \end{bmatrix}$

$A^5 =$ final $\begin{bmatrix} 0 & 1 & -3 & 2 & -4 \\ 3 & 0 & -4 & 1 & -1 \\ 7 & 4 & 0 & 5 & 3 \\ 2 & -1 & -5 & 0 & -2 \\ 8 & 5 & 1 & 6 & 0 \end{bmatrix}$

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Q.5) a)

	Action	visit order	stack (left → right, top at right)
1)	discover B	B	[B]
2)	discover C	B, C	[B, C]
3)	discover F	B, C, F	[B, C, F]
4)	discover H	B, C, F, H	[B, C, F, H]
5)	" G	B, C, F, H, G	[B, C, F, H, G]
6)	" D	B, C, F, H, G, D	[B, C, F, H, G, D]
7)	pop D	-	[B, C, F, H, G]
8)	pop G	-	[B, C, F, H]
9)	discover I	B, C, F, H, G, D, I	[B, C, F, H, I]
10)	pop I	-	[B, C, F, H]
11)	" H	-	[B, C, F]
12)	" F	-	[B, C]
13)	" C	-	[B]
14)	discover E	B, C, F, H, G, D, I, E	[B, E]
15)	discover A	B, C, F, H, G, D, I, E, A	[B, E, A]
16)	pop A	-	[B, E]
17)	pop E	-	[B]
18)	pop B	-	[]

order: B C F H G D I E A

b)

* tree edges:
  - ↳ B → C
  - ↳ B → E
  - ↳ C → F
  - ↳ F → H
  - ↳ H → G
  - ↳ H → I
  - ↳ G → D
  - ↳ E → A

* back edges
  - ↳ A → B
  - ↳ D → H
  - ↳ I → H

* front edges
  - ↳ F → I

* cross-edges
  - ↳ A → D
  - ↳ E → H

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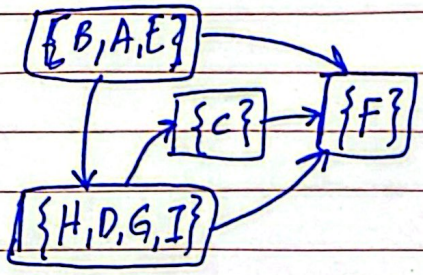
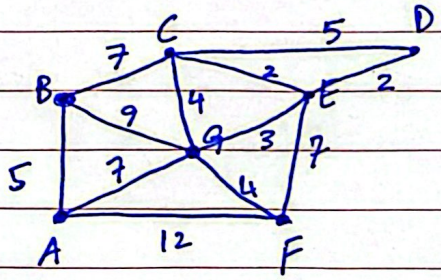
c) strongly connected components

↳ $S_0 = \{B, A, E\}$ ↳ $S_1 = \{C\}$ ↳ $S_2 = \{F\}$ ↳ $S_3 = \{H, D, G, I\}$



Q.6)

a)



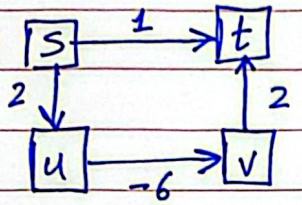
step	picked edge	S (after pick)	cut edges (next choices)	cumulative weight
0	—	{A}	A-F(12), A-M(7), A-B(5)	0
~~1~~	~~A-F(12)~~			
1	A-F(12)	{A,F}	A-M(7), A-B(5), F-E(7), F-M(7)	12
2	F-E(7)	{A,F,E}	A-M(7), E-M(3), E-D(2), A-B(5)	19
3	A-M(7)	{A,F,E,M}	M-B(9), M-C(4), M-F(4), A-B(5)	26
4	M-B(9)	{A,F,E,M,B}	B-C(7), M-C(4), A-B(5)	35
5	B-C(7)	{A,F,E,M,B,C}	C-D(5), C-E(2), M-C(4)	42
6	C-D(5)	{A,F,E,M,B,C,D}	—	47

cost = 47

day / date:

b) Yes, both can produce different maximum spanning trees (with different sets of edges), but the total weight will remain the same (both are optimal)

Q.7) a)



↳ According to Dijkstra algorithm would be 1 as from $s \to t = 1$. As that's the instant shortest edge. But when other edges are traversed, the cost will become negative weighted

$s \to u \to v \to t = -2$

s	t	u	v
s	(1,s)	(2,s)	(∞,-)
st		(2,s)	(∞,-)
stu			(-4,u)

b) No, adding a constant k, to all edges is not valid as adding k changes the cost of a path (n(edges) x k), so paths w/ differed lengths are distorted differently. So a path that was originally shorter, would become longer after addition of k. Hence the shortest-path ordering is not preserved.

c) Dijkstra works correctly only when negative edges leave the source node s, as all negative relaxations happen immediately during initialization (distance updates from s). After that, all remaining edges are non-negative, so the standard correctness proof of dijkstra applies.